



1 APPROACH — ONE CHAIN, FROM THE SHED TO THE STATE

Today a sick animal is spotted by a farmer who has no channel, while the State that needs the signal hears it weeks later on paper. We close that loop inside **one platform**: a symptom spoken in Marathi becomes a triaged case, a lab-confirmed outbreak, a detected cluster, an auto-issued task and a paid compensation claim — **with no re-entry at any step**. Reporting is made to *pay*, which is what actually fixes under-reporting.

 FARMER voice report • offline queue	 FIELD WORKER verify • photo • GPS	 VETERINARIAN triage queue • sample	 LABORATORY chain-of-custody • result	 BLOCK / DISTRICT cluster • task • claim	 STATE SITREP • forecast • policy
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2 PHASED ROLLOUT — EACH PHASE GATED ON A NUMBER, NOT A DATE

PHASE 0 • DONE Working Prototype - All six innovations built and demoable end-to-end 7 role dashboards, 53 REST endpoints 24 months of seeded epidemiology - Deployed on Render + Postgres, public HTTPS GATE ✓ full loop runs live in under 2 min	PHASE 1 • MONTH 1-3 Block Pilot - One block • ~50 villages • ~2 000 animals - Read-only sync with Bharat Pashudhan (BPID) - Train field workers; 1962 call-centre script - Vet sign-off on every triage output GATE ≥60% of reports filed by farmers, not staff	PHASE 2 • MONTH 4-9 District Rollout - One full district, all blocks + DDL laboratory - Two-way INAPH / LHDCP integration - Movement permits at checkpoints and markets - Compensation disbursed through DBT GATE detection lead-time beats paper by ≥7 days	PHASE 3 • MONTH 10-18 State Scale 36 districts on MahaIT / NIC cloud - Breed model on a GPU node; more languages - NADCP drive planning driven by the forecast - Open API for ICAR and One Health / IDSP GATE State SITREP replaces the manual return
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3 TECHNOLOGY, AND WHY EACH CHOICE

LAYER	CHOICE	WHY THIS, FOR THIS PROBLEM
Client	Vanilla-JS PWA	Installs on a ₹5 000 Android, works offline , no app-store gate
Server	FastAPI • SQLAlchemy	53 typed endpoints; auto OpenAPI for partner integration
Data	PostgreSQL / PostGIS	Spatial queries at State scale; SQLite for a zero-setup demo
Detection	Space-time scan	Kulldorff statistic — peer-reviewed , yields a real p-value, not a hunch
Forecast	SEIR on village graph	Answers “ring-vaccinate here — how many cases does that prevent?”
Vision	EfficientNetV2 • 51 breeds	Subject + quality gate rejects non-cattle instead of guessing
Assistant	Rules → Gemini 2.5	Rules act instantly and offline; the LLM takes free-form Hindi/Marathi
Reach	PWA + IVR 1962 + SMS	The farmer without a smartphone is still a sensor

4 RISKS WE DESIGNED AGAINST

RISK	MITIGATION ALREADY IN THE BUILD
Farmers under-report — no incentive	Report → claim → approved compensation . Reporting pays.
No network in the field	Offline-first PWA — the report queues locally and syncs on reconnect
No smartphone , low literacy	Marathi–Hindi voice, IVR on 1962, SMS, icon-first UI
False alarms erode trust	p-value threshold <i>plus</i> a vet-confirmation gate before escalation
Wrong AI diagnosis — liability	Emits <i>suspected category + severity + route</i> , never a diagnosis
Duplicate entry against BPID	API sync with Bharat Pashudhan; the tag ID is the shared key
GPU cost at scale	5 of 6 features need no model; vision degrades to manual entry, honestly
Officer adoption	Tasks are <i>auto-created</i> from clusters — the queue does the chasing

5 WHAT IS ALREADY BUILT, AND MEASURABLE TODAY

7 973 lines of production code	53 REST endpoints • 7 dashboards	17 diseases in a hot-reload KB	1 026 cases • 24 months • 87 locations
✓ Pashu Lens — 51-breed ID	✓ Outbreak Radar — scan statistic	✓ What-if ring-vaccination planner	✓ Health Passport + movement permit
✓ पशु मित्र voice assistant	✓ Compensation-claim loop	✓ Auto-generated SITREP	→ Next: BPID sync
			→ Next: DBT payout
			→ Next: GPU node for vision